EXPERIMENT NO: 10

Experiment to understand array function in Data science.

PROGRAM:

```
import numpy as np
array=np.random.randint(1,100,9)
array
```

OUTPUT:

```
array([58, 86,  1, 85, 18, 27, 68, 51, 41])
```

PROGRAM:

```
np.sqrt(array)
```

OUTPUT:

```
array([7.61577311, 9.2736185 , 1.        , 9.21954446, 4.24264069,
       5.19615242, 8.24621125, 7.14142843, 6.40312424])
```

PROGRAM:

```
Array.ndim
```

OUTPUT:

1

PROGRAM:

```
new_array=array.reshape(3,3)
new_array
```

OUTPUT:

```
array([[58, 86,  1],
       [85, 18, 27],
       [68, 51, 41]])
```

PROGRAM:

```
New_array.ndim
```

OUTPUT:

2

PROGRAM:

```
new_array.ravel()
```

OUTPUT:

```
array([83, 25, 19, 47, 62, 15, 96, 39, 51])
```

PROGRAM:

```
newm=new_array.reshape(3,3)
newm
```

OUTPUT:

```
array([[58, 86,  1],
       [85, 18, 27],
       [68, 51, 41]])
```

PROGRAM:

```
newm[2,1:3]
```

OUTPUT:

```
array([51, 41])
```

PROGRAM:

```
newm[1:2,1:3]
```

OUTPUT:

```
array([[18, 27]])
```

PROGRAM:

```
new_array[0:3,0:0]
```

OUTPUT:

```
array([], shape=(3, 0), dtype=int64)
```

PROGRAM:

```
new_array[0:2,0:1]
```

OUTPUT:

```
array([[58],
       [85]])
```

PROGRAM:

```
New_array[0:3,0:1]
```

OUTPUT:

```
array([[58],
       [85],
       [68]])
```

PROGRAM:

```
New_array[1:3]
```

OUTPUT:

```
array([[85, 18, 27],
       [68, 51, 41]])
```